Enhancing an Emotion-Labelled Video Dataset with Pose Estimation Key-points for Face, Body and Hands

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Abstract

Research in automated emotion recognition from video has traditionally focused on facial expression as the primary indicator of emotional state. More recently, increasing attention has been given to incorporating features from body posture and gesture alongside facial expression to improve the performance of emotion recognition models. Although many emotion-labelled video datasets exist, few incorporate pose estimation data, and many of those that do are collected in lab-based environments.

To address this limitation, we introduce a new emotion-labelled pose estimation dataset derived from the Multimodal EmotionLines Dataset (MELD). This dataset combines MELD's emotion annotations with speaker-specific, detailed pose estimation key-points from face, body and hands, extracted from the corresponding video recordings. The MELD-Keypoints v0.1 release contains pose estimation data extracted from 4,059 MELD video clips and 283,093 frame-level keypoint records. To date, 5,601 video clips have been processed through the complete extraction pipeline, of which 72.5% successfully produced speaker-specific pose data.

Keywords: Emotion Recognition, Computer Vision, Pose Estimation.

1 Introduction

Automated emotion recognition is a rapidly growing area of research as seen in the increased number of papers published on the topic since 2012 [Pan et al., 2023] with many applications including health and social care, education, sentiment analysis and human computer interaction [Pan et al., 2023, Pereira et al., 2024]. In particular there is opportunity to support the care and treatment of people living with conditions such as Parkinson's disease or dementia, in a care home environment where changes in emotional state can be a symptom of the disease and a reaction to medication changes; in such cases an automated emotion recognition tool would be useful for monitoring emotional wellbeing over time [Song, 2024].

Humans express emotion through audio, visual, physiological and textual signals and correspondingly there are various methods for observing these different signals [Pereira et al., 2024]; taking inspiration from how we as humans observe emotion in others this study focuses on visual signals as expressed in video. There are many emotion-labelled video datasets available, some that have been specifically designed for that purpose [Bänziger et al., 2012, Busso et al., 2008, Miranda-Correa et al., 2021, Park et al., 2020], and others which have been created from existing material like movies or tv shows and adapted for emotion research [Poria et al., 2019, Dhall et al., 2012 and Jiang et al., 2020]. A limitation that we observed in some of the lab-based video datasets was that there was minimal furniture and often only a single participant, potentially reducing the ability of trained models to generalise to real-world setting. The datasets which used clips from film or television, are acted datasets but the settings are usually in real-world type environments, with the challenges that can bring, such as movement around and interactions with other people and objects in the environment.

The main contribution of this paper is MELD-Keypoints, a novel speaker-specific pose estimation extension of the Multimodal EmotionLines dataset (MELD). By combining OpenPose pose estimation with pyppbox-based identity tracking, the dataset augments the MELD emotion annotations with detailed key-point data for face, body and hands, for the target speaker associated with each utterance. This provides a new resource for investigating multimodal emotion recognition in complex, real-world, multi-person environments.

2 Related Work

Historically, emotion recognition research has focused on text analysis, audio signal analysis, physiological signals, and in the context of image and video data, primarily on using facial expressions to determine emotion [Lu et al., 2024]. Data for physiological signals, such as EEG, heart rate and temperature, requires the use of a wearable device which often has to touch the skin; the user must remember to wear it and if they find it uncomfortable they may not be diligent about doing so [Cai et al., 2023]. The benefit of video data is that it can gather information for everyone onscreen including potentially valuable context information about the environment and interactions taking place with others; however issues with privacy are a concern as are the challenges of missing data that will occur as a result of occlusion as people move around others and furniture or turn away from the camera, as well as people moving in and out of the camera field of view [Geetha et al., 2024].

Whilst emotion recognition research using image or video data has mainly focused on facial expressions, there has been a shift to skeleton-based emotion recognition [Lu et al., 2024]. Humans communicate their emotions via facial expressions, and also via their body posture and hand gestures, therefore capturing this type of information has the potential to improve emotion categorisation [Lu et al., 2024]. This shift has been aided by the variety of methods to obtain skeleton data, whether by using motion capture [Busso et al., 2008], a depth-sensing camera like Kinect [Zhang, 2012 and Li et al., 2016] or applying pose estimation tools to RGB video to extract body pose key-points, such as OpenPose [Cao et al., 2021, Raaj et al., 2019, Simon et al., 2017], AlphaPose [Fang et al., 2023], MediaPipe [Bazarevsky et al., 2020].

There have been several models created to define emotions and most fall into one of two types: categorical or dimensional [Younis et al., 2024]. One of the most popular categorical models known as the Ekman model [Ekman and Friesen, 2003], focuses on six main emotions: anger, disgust, fear, happiness, sadness and surprise. This model is frequently used in emotion recognition datasets such as MELD [Poria et al., 2019], AFEW [Dhall et al., 2012] and DFEW [Jiang et al., 2020], usually with a seventh category of neutral.

Several emotion-labelled datasets including GEMEP, IEMOCAP, AMIGOS and K-EmoCon were collected in controlled laboratory environments with limited environmental complexity and participant movement [Bänziger et al., 2012, Busso et al., 2008, Miranda-Correa et al., 2021, Park et al., 2020]. In contrast, AFEW and DFEW were created from film and television content to support the research of emotion in more natural, real-world type environments [Dhall et al., 2012, Jiang et al., 2020]. MELD – Multimodal EmotionLines Datasets [Poria et al., 2019], is a similar 'in-the-wild' dataset which contains clips from the tv show 'Friends'. A benefit of this dataset is that the main 6 characters appear in the majority of the videos, which will provide a large variety of emotion data for each of the 6 characters. Many clips have more than one person in the frame and there is movement of people around other people or furniture or turning away from the camera, providing instances of occlusion which is something we would expect tackle in real-world situations.

Most of these datasets are RGB video datasets and would, therefore, need to be processed through a pose estimation tool in order to extract key-point data. Two exceptions are, IEMOCAP – the Interactive Emotional Dyadic Motion Capture Database [Busso et al., 2008], which contains motion-capture data and detailed facial markers, and EMILYA [Fourati and Pelachaud, 2014] which focuses on bodily emotion expression and did not include facial expression data. Both datasets were collected in laboratory environments and do not provide the combination of natural multi-person interactions and detailed face, body and hand key-points targeted by this work.

3 Methodology

In this section we will introduce the dataset that we have chosen to augment with pose estimation key-points for face, body and hands and the tools used to accomplish this task.

3.1 MELD Dataset

For this study we selected Multimodal EmotionLines Dataset (MELD) [Poria et al., 2019], a video dataset containing 13,848 video clips from the TV show Friends. MELD is an extension of EmotionLines [Hsu et al., 2018], which was created using the scripts from 'Friends' which had been divided into sequences of utterances, with each utterance assigned an emotion label, using the Ekman model [Ekman and Friesen, 2003] of 7 emotion categories plus neutral and a sentiment label (positive, negative, neutral) making this dataset useful for sentiment analysis and emotion categorisation from text.

The team that created MELD identified an opportunity to enhance and extend the original EmotionLines dataset by including video and audio to enable researchers to explore emotion recognition from visual and audio signals as well as text, with potential to combine these signals for multi-modal emotion recognition. To this end the emotion labelling for each utterance was repeated, with a new set of reviewers who read the text, watched the video and listened to the audio, before making their assessment. This new set of reviewers who had the benefit of visual and audio signals did in some cases make different decisions to the prior group who relied solely on the text.

The MELD dataset contains the video clips of each utterance, which is identified by a video ID, a dialog ID, an utterance ID, the name of the person who is saying the utterance, the text of the utterance and an emotion label and sentiment label. In our augmented dataset we have used the emotion label rather than the sentiment label as for the purposes of our project, we were more interested in changes in emotion, particularly when applied to a care home setting where we would be tracking emotion changes of residents over time.

3.2 OpenPose

OpenPose is a 2D pose estimation tool which can detect up to 135 key-points for multiple people and is capable of real-time processing [Cao et al., 2021, Raaj et al., 2019, Simon et al., 2017]. It takes a bottom-up approach, unlike tools such as YOLO [Redmon et al., 2016], which first detect people in frame before finding the key-points, OpenPose first finds the key-points by using heatmaps to identify body joints (elbows, wrists, knees etc.) and then Part Affinity Fields (PAFs) to create vector maps to link the different body joints, before finally pairing joints together to create the skeleton.

We selected OpenPose because of its three comprehensive key-point models for face, body and hand which provided a significant amount of pose estimation data as input to features for emotion recognition models. We also chose it for its performance in being able to provide body pose key-points for multiple people on-screen, which is the case for many of the videos within the MELD dataset.

However, as the emotion label for each video in MELD is only connected to the person who says the utterance in that video, and there are often multiple people onscreen, pyppbox was selected to identify, track and extract only the key-points for the utterance speaker in each frame.

3.3 pyppbox

Pyppbox is an open-source tool that combines several other open-source tools within the PoseTreID framework for person detection across multiple frames. [Siv et al., 2020, Siv et al., 2022]. For the Identification module we used FaceNet, a facial recognition model, which we pre-trained to recognize the six main characters from Friends, using an image dataset from Kaggle [Kalabasi, 2022].

Due to the changes made, the tool was able to process videos and provide an output that contains the frame number, the people identified in that frame, an identity label each person, and a bounding box indicating the location of that identified person within the frame.

3.4 Creating a New Augmented Dataset

The end-to-end process used to create this extension of the MELD dataset with the existing data augmented with pose estimation key-points is illustrated in Figure 1. The MELD video dataset is processed through both OpenPose and pyppbox. OpenPose returns a folder of JSON files for each video, containing a JSON file for every frame and hand, body and face key-points for each person it detects in the frame. Pyppbox, whose reidentification model FaceNet has been pre-trained to recognise the six main Friends characters, will return a text file for each video that contains an entry for each person it detects in each frame, an identify label for that person and bounding box and mini-frame coordinates.

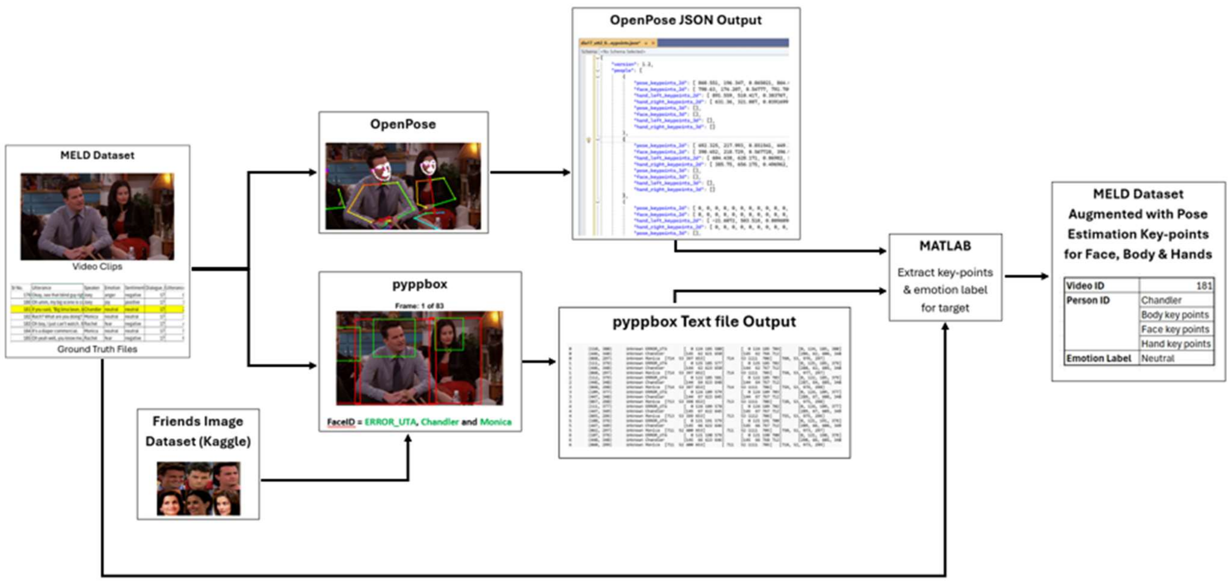

Figure 1: End-to-End Process to create new version of MELD dataset augmented with detailed pose estimation key-points

Combining the Ground Truth files from MELD, the OpenPose output JSON files and the pyppbox output text files; for each video it identifies from the MELD ground truth file the target person who has said the utterance and extract the bounding box coordinates for that person, for every frame from the relevant pyppbox output text file. Then accessing the corresponding OpenPose JSON file, for each frame, the OpenPose body key-points were used to construct a bounding box for each detected individual. These candidate bounding boxes were compared with the pyppbox bound box corresponding to the target speaker. Matching was performed using a combined score based on Intersection over Union (IoU) and the proportion of valid pose key-points located within the pyppbox bounding box. The highest-scoring OpenPose detection was selected and the corresponding face, body and hand key-points extracted. This is then collated into the new dataset that contains the video ID, the identity of the target person and the face, body and hand key-points for that person in every frame of the video and the emotion label which was also obtained from the MELD ground truth file.

4 Results

All 13,848 MELD video clips have been processed through OpenPose, with a single failure caused by a corrupted video clip. To date 5,607 video clips have been processed through pyppbox, again with the same single failure. Combining the available OpenPose and pyppbox outputs produced 5,601 successfully processed videos, from which 4,059 speaker-specific videos were added to the MELD-Keypoints v0.1 dataset, representing 72.5% of the processed videos. The remaining 1,542 videos were excluded because the target speaker was not one of the six principal characters, could not be identified by pyppbox, or due to missing data. All exclusions were recorded in tracking tables for subsequent analysis.

Split	MELD Videos	OpenPose Outputs	Pyppbox Outputs	MATLAB Processed	Videos Added to New Dataset
Dev	1,112	1,112	1,112	1,111	829
Test	2,747	2,747	1,000	1,000	732
Train	9,989	9,988	3,495	3,490	2,498
Total	**13,848**	**13,847**	**5,607**	**5,601**	**4,059**

Table 1: Progress of new dataset generation

The current release is MELD-Keypoints v0.1 and the link to download this version of the dataset and associated MATLAB processing framework, is publicly available through a GitHub Repository[1].

We have carried out an initial analysis of the balance of classes and character distribution of the current dataset.

Emotion	Count	%
Neutral	1832	45.10%
Joy	632	15.60%
Anger	538	13.30%
Surprise	503	12.40%
Sadness	296	7.30%
Fear	132	3.30%
Disgust	126	3.10%
Total	**4059**	**100%**

Table 2: Emotion Label Distribution

Character	Count	%
Ross	785	19.30%
Joey	728	17.90%
Phoebe	663	16.30%
Chandler	653	16.10%
Monica	636	15.70%
Rachel	594	14.60%
Total	**4059**	**100%**

Table 3: Character Distribution

5 Discussion

The current MELD-Keypoints v0.1 release demonstrates that large-scale speaker-specific key-point extraction from MELD is feasible. The extraction pipeline successfully generated pose data for 4,059 videos and 283,093 frame-level records, corresponding to 72.5% of the videos processed through the complete pipeline.

Whilst there are other emotion recognition video datasets with accompanying pose estimation data available, these have been typically lab-based with acted out emotions [Busso et al., 2008]. However MELD, whilst being an acted dataset, is set in real-world type environments like living rooms, coffee shops and offices, with people moving around furniture and other people, similar to what we would expect to observe in a true real-world dataset, such as in a care home setting. The occlusion which occurs when one person moves in front of another or turns away from the camera is something we are keen to investigate and is a benefit of this dataset, firstly because these are typical scenarios in the original MELD dataset, but also that the OpenPose has shown good performance at handling occlusions and the reidentification and tracking elements of pyppbox have ensured that even when a person turns away from the camera, if they have previously been identified, then they continue to be tracked and linked to that identify and the key-point data that is still visible can be extracted, for example we may be missing some or all of the face key-points when they turn away but we still have body and hand key-points. This gives us scope to investigate how to handle occlusion in an emotion recognition system.

OpenPose was selected for its comprehensive key-point models for face, body and hand and its ability to return results for multiple people and handle occlusions well, however when we have carried out the next stage of our study to consider which combination of features from face, body and hands works best for emotion recognition,

[1] https://github.com/CSempey/MELD-Keypoints

this will allow us to assess if we need this level of key-point detail and whether an alternative, less detailed pose estimation model would be sufficient.

Although the scenarios are acted the emotions expressed are recognisable, though we would anticipate that any emotion recognition models trained on this data would be fine-tuned on datasets similar to the type the models would be used on, for example, care home video data.

A major factor limiting dataset coverage is that the FaceNet model used within pyppbox was trained only on the six principal Friends characters. Consequently, pose data can only be extracted when the target speaker belongs to this group. Although as we saw in MELD-Keypoints v0.1, data for these 6 main characters has accounted for 72.5% of the data processed so far. Expanding the facial recognition model to additional recurring characters is expected to increase coverage in future releases.

Looking further ahead, once we have created a suitable emotion recognition model, as we already have OpenPose key-point data for everyone in the video, there is an opportunity use the model to expand the dataset to incorporate emotion labels for everyone onscreen, enabling further research into emotion within groups. It is even possible to combine the data for sequence of videos that belong to the same dialog, so that you could look at emotion changes during conversation or over a longer period of time than just an utterance.

6 Conclusions

In this paper we have highlighted the need for large-scale, good quality emotion-labelled video datasets augmented with pose estimation data, set in real-world type environments with multiple people interacting and moving around other people and furniture, causing occlusion, moving in and out of camera view; this would support the development of emotion recognition models that could be more readily applied to a real-world environment. To address this need, we introduced MELD-Keypoints, a speaker-specific extension of the MELD video dataset that augments existing emotion annotations with face, body and hand key-points extracted using OpenPose and linked to identified speakers using pyppbox. The current MELD-Keypoints v0.1 release contains pose data for 4,059 videos and 283,093 frame-level key-point records. By providing emotion labels together with detailed facial, body and hand motion data in complex social scenes, MELD-Keypoints represents a valuable resource for future multimodal emotion recognition research

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